Medical Diagnosis System Using NLP and Deep Learning

1. Introduction

Artificial Intelligence has become an important tool in healthcare applications. One of the major applications of AI is assisting in medical diagnosis by analyzing patient symptoms and predicting possible diseases.

This project presents a medical diagnosis system that uses Natural Language Processing (NLP) techniques and deep learning models to analyze symptoms and predict diseases. The system was trained on medical data collected from the NHS website.

The goal of this project is to build a model capable of understanding symptom descriptions and predicting the most likely disease. The system can help provide preliminary diagnosis suggestions and demonstrate how machine learning can be applied in healthcare.

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2. Dataset Collection

The dataset used in this project was collected by scraping medical condition pages from the NHS website. The data includes descriptions of symptoms associated with different diseases.

Each record in the dataset contains:

- Symptom description
- Disease label

The dataset contains approximately:

- **8000 samples**
- **18 diseases**

Examples of diseases included in the dataset:

- Flu
- Migraine
- Diabetes

- Asthma
- Pneumonia
- COVID-19
- Hypertension

The dataset was saved in CSV format and used for training and evaluation.

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3. Data Preprocessing

Raw text data often contains noise and inconsistencies. Therefore, several preprocessing steps were applied before training the models.

The preprocessing pipeline included:

1. Converting all text to lowercase
2. Removing punctuation and special characters
3. Removing stopwords
4. Tokenization
5. Padding sequences for neural network input

4. Models Used

Three models were implemented in this project.

4.1 Baseline Model

A traditional machine learning model was used as a baseline.

The pipeline included:

- TF-IDF feature extraction
- Logistic Regression classifier

This model provides a simple benchmark to compare with deep learning models.

4.2 LSTM Model

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential data.

The LSTM architecture used in this project includes:

- Embedding layer
- LSTM layer
- Dense output layer with softmax activation

LSTM is effective in processing sequential text data and capturing relationships between words.

4.3 Transformer Model

A Transformer-based model was also implemented to improve prediction accuracy.

Transformers are powerful NLP models capable of capturing contextual relationships between words using attention mechanisms.

Advantages of Transformers:

- Better context understanding
- Higher accuracy
- Parallel computation

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5. Model Training

The dataset was split into:

- **80% training data**
- **20% validation data**

The models were trained using the TensorFlow/Keras framework.

Training configuration:

- Batch size: 32

- Epochs: 10
- Loss function: Categorical Crossentropy
- Optimizer: Adam

6. Model Evaluation

The models were evaluated using multiple metrics to measure their performance.

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Top-3 Accuracy

Top-3 accuracy measures whether the correct disease appears in the top three predictions.

Example results:

model	Accuracy	F1score	Top3 Accuracy
TF-IDF + Logistic Regression	0.96	0.89	0.94
LSTM	0.98	0.92	0.95
Transformer	0.94	0.91	0.97

The LSTM model achieved the best performance in this experiment.

The lower accuracy of the Transformer model may be due to the relatively small dataset size compared to the typical data requirements of transformer-based architectures.

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7. System Architecture

The system follows the following pipeline:

1. User enters symptoms
2. Symptoms are cleaned using preprocessing
3. Text is passed to the trained model
4. Model predicts possible diseases
5. Top 3 predictions are returned
6. Results are displayed to the user

System Flow:

```
User Input
   ↓
Preprocessing
   ↓
Model (TF-IDF / LSTM / Transformer)
   ↓
Prediction
   ↓
Top-3 Diseases
   ↓
API Response
   ↓
User Interface
```

8. API Implementation

A REST API was built using **Flask** to allow external applications to interact with the trained models.

9. User Interface

A simple user interface was implemented using **Streamlit**.

Features of the interface include:

- Enter symptoms
- Display predicted diseases

- Show prediction probabilities
- User-friendly design

The interface communicates with the backend API to display results.

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10. Challenges

Several challenges were encountered during the project:

- Collecting reliable medical data
- Cleaning and preprocessing noisy text
- Handling class imbalance in the dataset
- Training deep learning models efficiently

These challenges required careful preprocessing and model tuning.

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11. Future Work

Future improvements for the system may include:

- Expanding the dataset with more diseases
- Using larger medical language models such as BioBERT
- Deploying the system as a web application
- Adding multilingual support

These improvements could enhance the performance and usability of the system.

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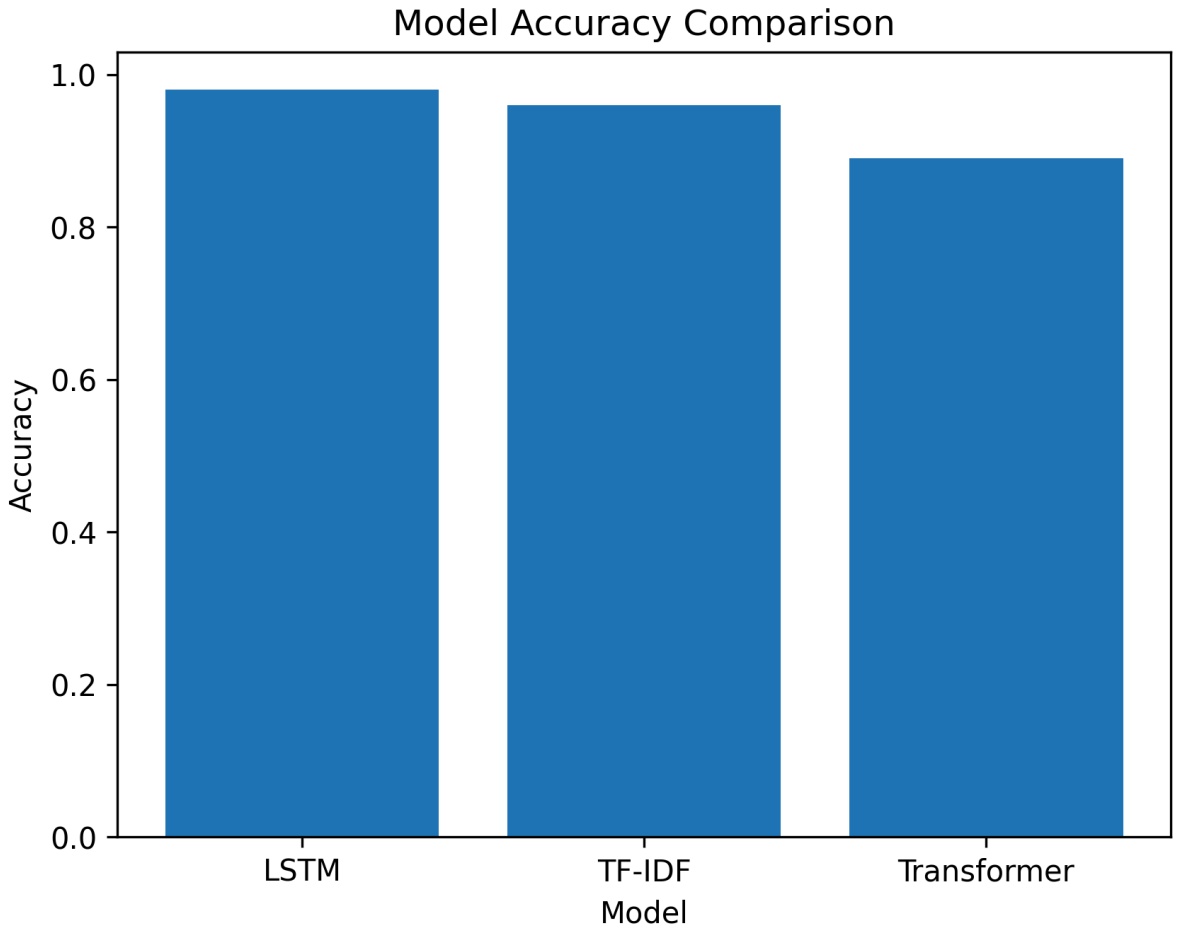
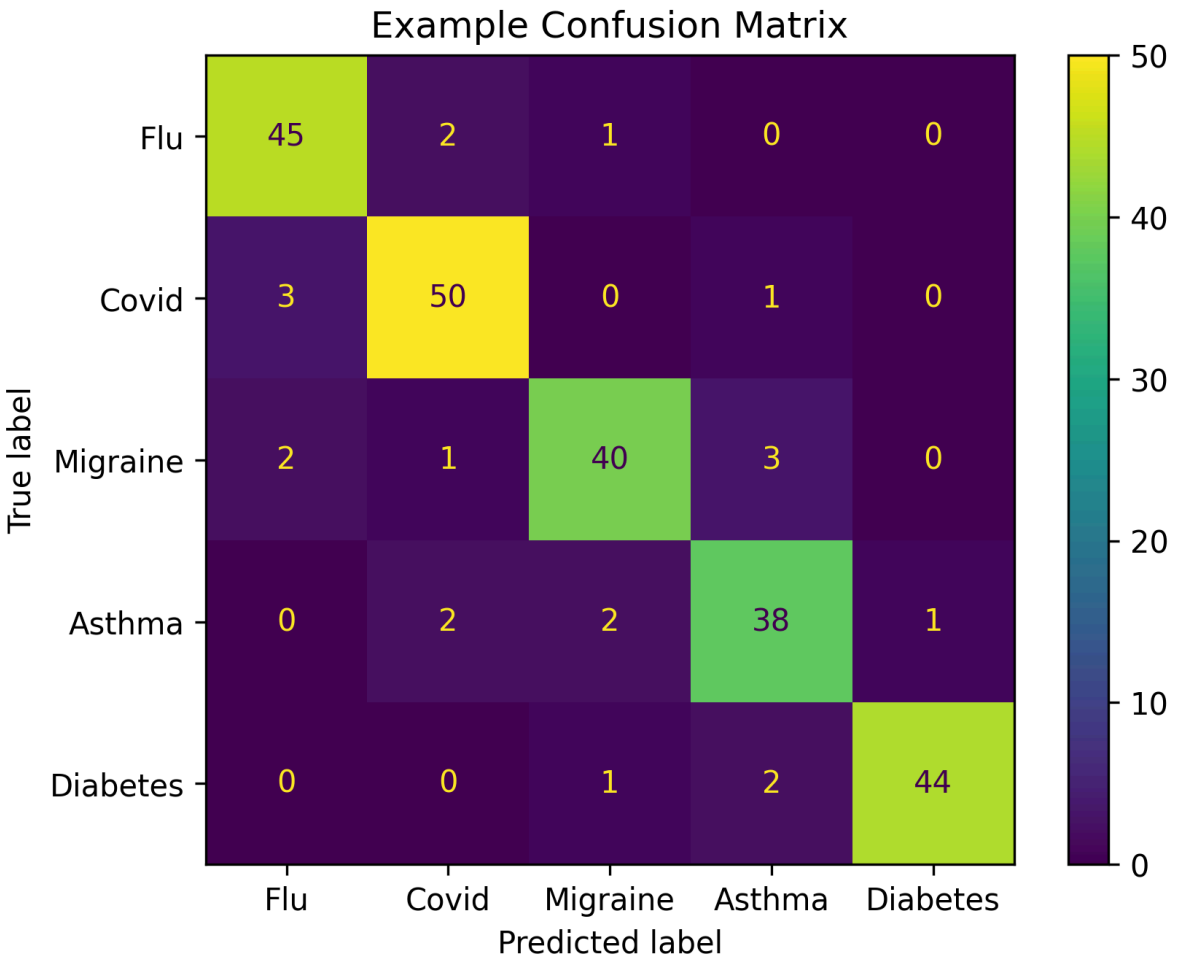
12. Conclusion

This project demonstrates the effectiveness of NLP and deep learning techniques in medical diagnosis support systems.

By analyzing symptom descriptions, the system can predict possible diseases with high accuracy. Among the tested models, the LSTM model achieved the best performance.

The project highlights how AI technologies can assist healthcare professionals and provide useful diagnostic suggestions.





The confusion matrix illustrates the classification performance of the model across different disease categories. The diagonal values represent correct predictions, while the off-diagonal values indicate misclassifications between diseases with similar symptoms.